

Predictive Lead Conversion Model - Machine Learning

Design and Implemented by – Akeem Asiru

Domain: EdTech | **Tech Stack:** Python, Scikit-learn, Pandas, Matplotlib, Seaborn

Models Used: Decision Tree Classifier, Random Forest Classifier

Objective: Identify potential leads likely to convert to paying customers

Objective

The project aimed to help ExtraaLearn—a growing EdTech startup—optimize its lead nurturing process by building a predictive model. The goal was to:

- Predict whether a lead will convert into a paid customer.
- Understand the key factors influencing lead conversion.
- Provide actionable insights for marketing and sales strategy.

Dataset Overview

- **Total records:** 4612 leads
- **Target variable:** status (0: not converted, 1: converted)
- **Key features:**
 - Demographics: age, current_occupation
 - Engagement metrics: time_spent_on_website, website_visits, page_views_per_visit
 - Interaction details: last_activity, first_interaction, profile_completed
 - Media exposure: referral, digital_media, print_media, etc.

Data Preprocessing & EDA

- **Handling Categorical Variables:** Used `pd.get_dummies()` for one-hot encoding.
- **Dropped** irrelevant fields like ID (unique per record).
- **Class Imbalance:** ~30% conversion rate; addressed via `class_weight` tuning.
- **Skewed Features:**
 - `website_visits` and `page_views_per_visit` showed right skewness.

- **Key Observations:**
 - Higher conversion rates for professionals and those who completed their profile.
 - Leads who interacted via **website** converted more than mobile app users.
 - Referral leads had high conversion rates despite being few in number.

Correlation Analysis

- Strongest positive correlations with conversion:
 - `time_spent_on_website`
 - `profile_completed_High`
 - `first_interaction_Website`
- Near-zero correlation: media features like print, digital media, and magazine exposure.

Modeling & Evaluation

1. Decision Tree Classifier (Default)

- **Training Accuracy:** 100% (Overfitting)
- **Test Accuracy:** ~81.3%
- **Recall for Class 1 (Converted):** ~70%

2. Decision Tree (Tuned via GridSearchCV)

- Parameters tuned: `max_depth`, `min_samples_leaf`, `criterion`
- **Test Accuracy:** Maintained ~81%
- **Interpretability:** Visualized decision paths
- **Top Features:**
 - `time_spent_on_website`
 - `first_interaction`
 - `profile_completed`

3. Random Forest Classifier

- **Training Accuracy:** 100% (overfit)
- **Test Accuracy:** ~85.1%
- **Recall (Class 1):** ~69%
- Feature importance spread across more variables → better generalization

4. Random Forest (Tuned)

- Parameters: `n_estimators`, `max_depth`, `min_samples_leaf`, `class_weight`, `max_samples`, `max_features`
- Best Recall Achieved:** ~78% with balanced performance
- Avoided Overfitting:** Reduced complexity improved generalization

Feature Importance (Top Drivers)

Feature	Importance
Time Spent on Website	34.8%
First Interaction (Website)	32.7%
Profile Completion (Medium)	23.9%
Age	6.3%

Business Impact & Recommendations

- Leads spending more time on the website and completing profiles are far more likely to convert.
- Prioritize outreach to **professional** and **unemployed** leads over **students**.
- Boost **referral programs**, as they show high conversion potential.
- Improve mobile app user experience to increase conversions through that channel.
- Integrate this model in CRM to rank and score leads in real time.

Tools & Techniques Used

- Python libraries:** pandas, numpy, seaborn, matplotlib, sklearn
- Algorithms:** Decision Tree, Random Forest, GridSearchCV for hyperparameter tuning
- Evaluation Metrics:** Accuracy, Precision, Recall, F1-Score, Confusion Matrix
- Scoring Metric:** Recall prioritized to minimize false negatives

Final Model Performance Summary

Model	Train Accuracy	Test Accuracy	Recall (Class 1)
Decision Tree	100%	81%	70%
Tuned Decision Tree	~100%	81%	~70%
Random Forest	100%	85%	69%
Tuned Random Forest	88%	85%+	78%

Conclusion

This project successfully delivered a lead conversion prediction model with real business value. It highlights the importance of behavioral signals (time spent, profile completeness) and showcases how AI can directly support sales optimization. With recall-focused evaluation, the solution aims to **maximize potential customer capture** while minimizing wasted outreach.